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ENGINEERING



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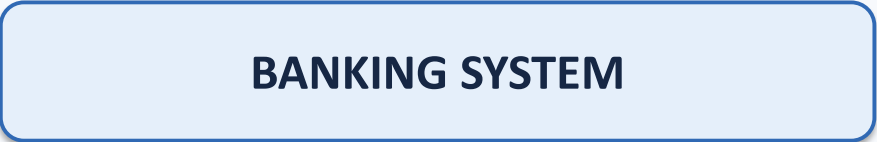
CSA0408 - Operating Systems

Operating System for High-Volume Banking Transactions

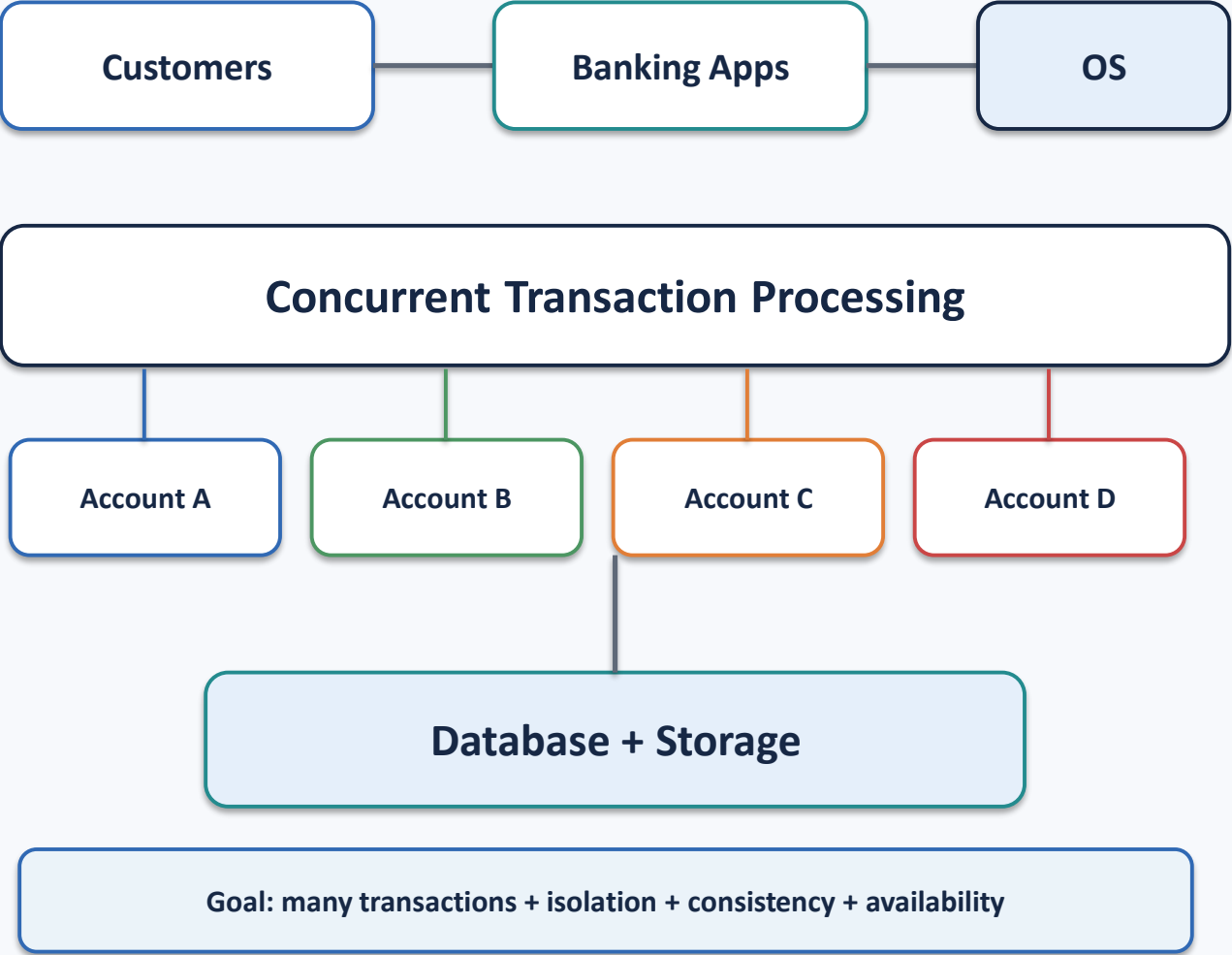
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Operating System for High-Volume Banking Transactions

Types of OS, OS structure and system calls

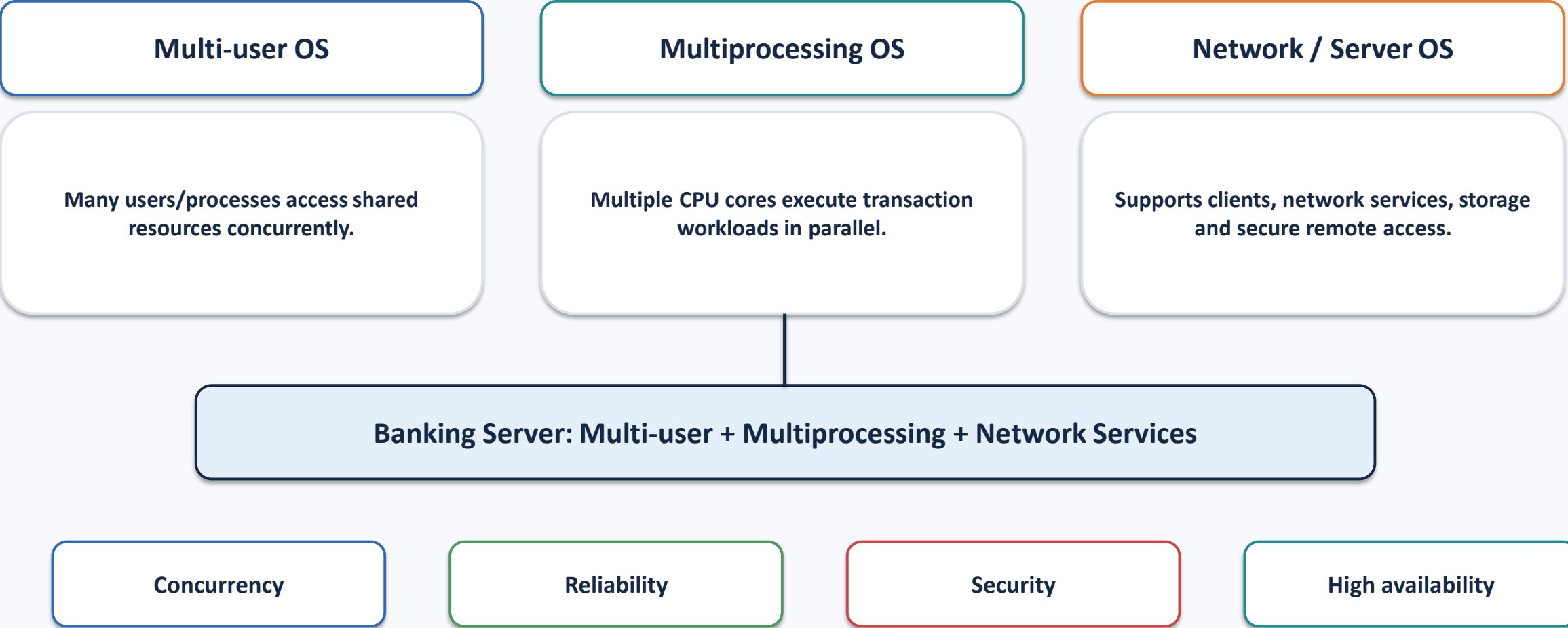


- Thousands of customers may initiate transactions simultaneously.
- OS must support concurrency, security, reliability and fast I/O.
- Transactions include deposit, withdrawal, transfer and balance enquiry.



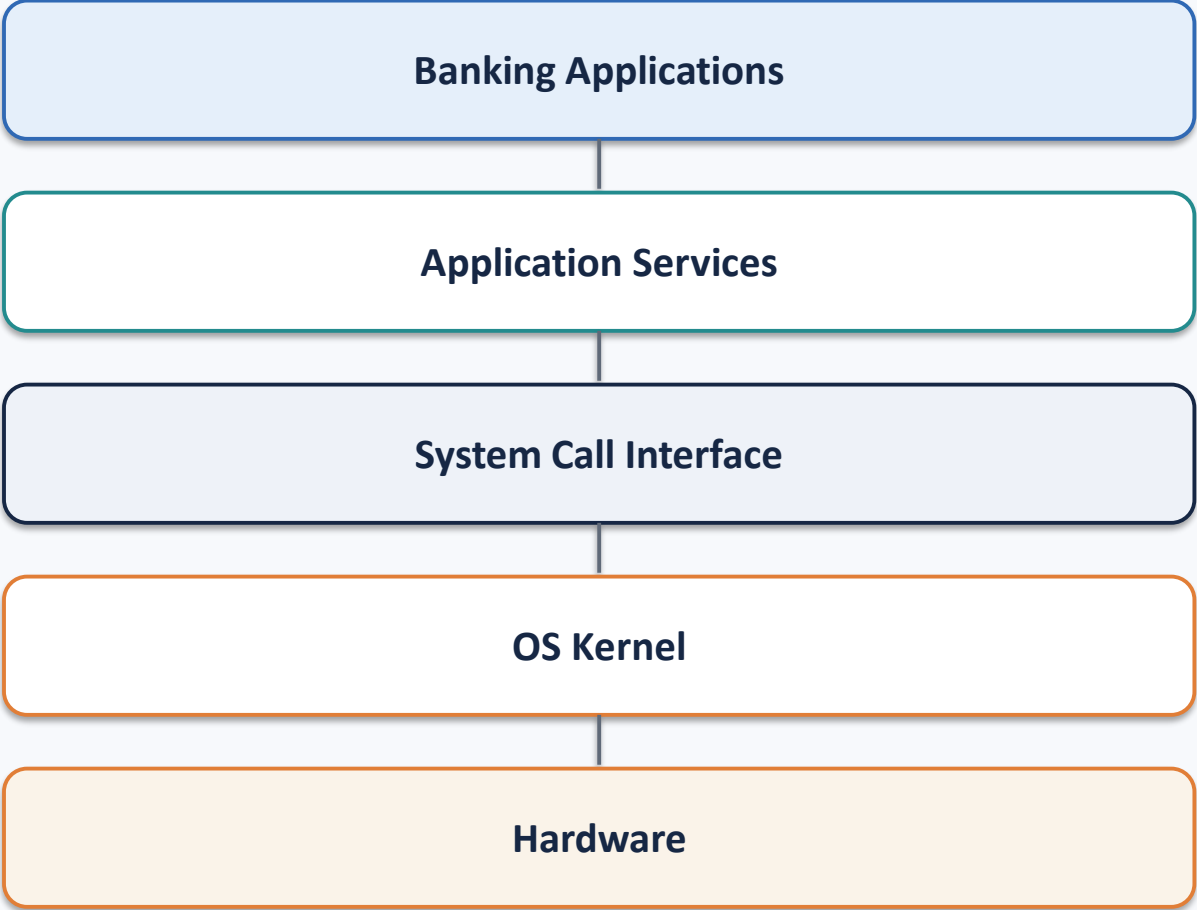
1. Types of OS Suitable for Banking

Different OS characteristics support different parts of a banking workload



2. OS Structure Used in a Banking Server

Layered / modular architecture connects banking services to protected kernel resources



ATM • Mobile Banking • Internet Banking

Authentication • Transaction Manager • Audit

Controlled entry to kernel services

Process • Memory • File • I/O • Network • Security

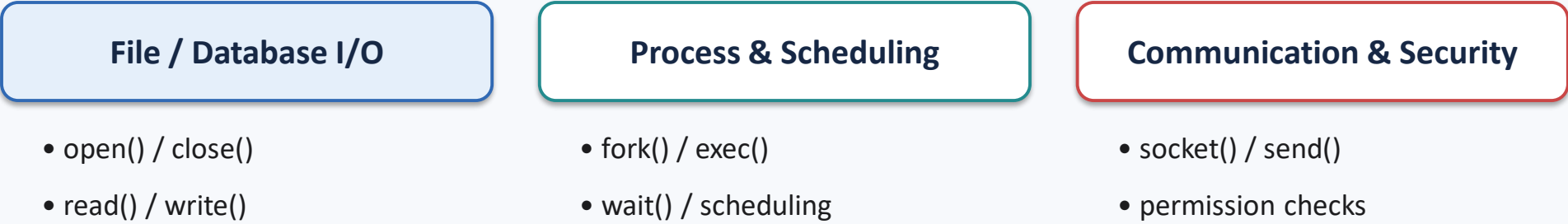
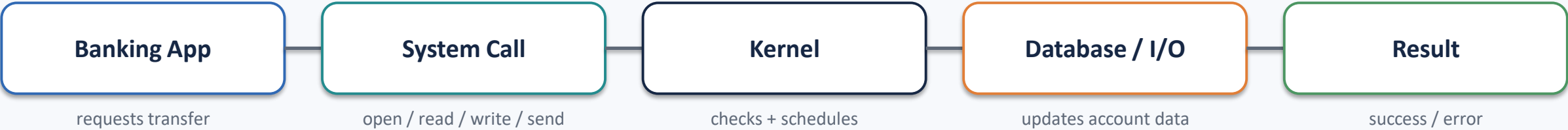
CPU • RAM • Disk • Network Interface

Security + Isolation

Logging + Recovery

3. Role of System Calls in Transaction Processing

System calls provide the controlled bridge between banking software and the OS kernel



4. Complete Transaction Processing Flow

Concurrency control, system calls and kernel services work together for safe execution



Thousands of Transactions → Scheduling + Memory Protection + I/O + Synchronization

Consistency

Correct account balances

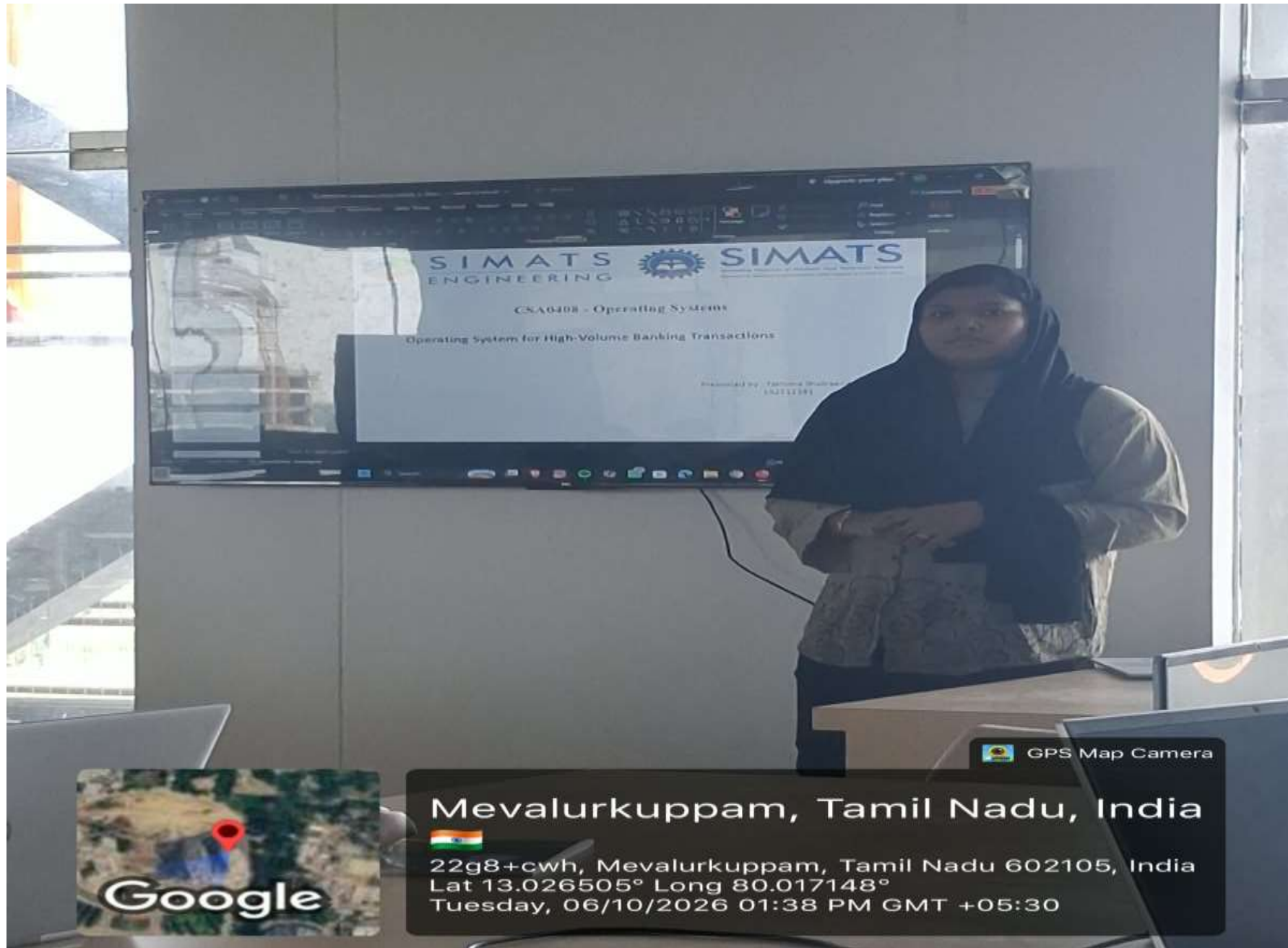
Isolation

Transactions do not corrupt each other

Availability

Service continues under heavy load

Key takeaway: The OS enables secure, concurrent and reliable transaction processing.



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